전산화학 (2013년 1학기) **1차 시험 해답**

(총 5 문제, 60 점)

1. **(총 20점)**

(a) **(5점)**

$$z = \frac{3+2i}{3-2i} = \frac{3+2i}{3-2i} \times \frac{3+2i}{3+2i} = \frac{5+12i}{13} = \frac{5}{13} + \frac{12}{13}i$$

$\therefore \mathrm{Re}(z) = \frac{5}{13}, \qquad \mathrm{Im}(z) = \frac{12}{13}$

(b) **(5점)** $1+\sqrt{3}i = 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 2\big(\cos(\pi/3) + i\sin(\pi/3)\big) = 2e^{i\pi/3}$ 이므로

$$\begin{aligned} z = (1+\sqrt{3}i)^{50}e^{-i\pi/2} &= \big(2e^{i\pi/3}\big)^{50}e^{-i\pi/2} \\ &= 2^{50}e^{i50\pi/3}e^{-i\pi/2} \\ &= 2^{50}(e^{i48\pi/3}e^{i2\pi/3})e^{-i\pi/2} \\ &= 2^{50}e^{i2\pi/3}e^{-i\pi/2} \\ &= 2^{50}e^{i(2\pi/3-\pi/2)} \\ &= 2^{50}e^{i\pi/6} \\ &= 2^{50}\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right) \\ &= 2^{50}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = 2^{49}\sqrt{3} + 2^{49}i \end{aligned}$$

$\therefore \mathrm{Re}(z) = 2^{49}\sqrt{3}, \qquad \mathrm{Im}(z) = 2^{49}$ 이다.

(c) **(10점)** Euler 공식으로부터

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{1a}$$

$$e^{-i\theta} = \cos\theta - i\sin\theta, \tag{1b}$$

이므로, (1a) + (1b) $\Rightarrow 2\cos\theta = e^{i\theta} + e^{-i\theta} \Rightarrow \boxed{\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}}$

$$\begin{aligned} z = \cos(a+ib) &= \frac{e^{i(a+ib)} + e^{-i(a+ib)}}{2} \\ &= \frac{e^{ia-b} + e^{-ia+b}}{2} \\ &= \frac{1}{2}\big(e^{-b}e^{ia} + e^{b}e^{-ia}\big) \\ &= \frac{1}{2}\big[e^{-b}\cos a + ie^{-b}\sin a + e^{b}\cos(-a) + ie^{b}\sin(-a)\big] \\ &= \frac{e^{-b}\cos a}{2} + \frac{e^{b}\cos a}{2} + i\left[\frac{e^{-b}\sin a}{2} - \frac{e^{b}\sin a}{2}\right] \end{aligned}$$

$\therefore \mathrm{Re}(z) = \frac{e^{-b}\cos a}{2} + \frac{e^{b}\cos a}{2}, \qquad \mathrm{Im}(z) = \frac{e^{-b}\sin a}{2} - \frac{e^{b}\sin a}{2}$

2. **(총 10점, 각 5점)**

(a) $z = \sqrt{3} + i$ 이므로 $r = \sqrt{(\sqrt{3})^2 + 1^2} = 2$ 이고, $\tan\theta = 1/\sqrt{3} \Rightarrow \theta = \pi/6$

그러므로 $z = \sqrt{3} + i = 2e^{i\pi/6}$

(b) $z = -2 - \sqrt{12}i$ 이므로 $r = \sqrt{(-2)^2 + (-\sqrt{12})^2} = 4$ 이고, $\tan\theta = \sqrt{3} \Rightarrow \theta = 4\pi/3$
(z가 3사분면에 있으므로)
그러므로 $z = \sqrt{3} + i = 4e^{i4\pi/3}$

3. **(총 10점, 각 5점)**

(a) **(5점)**

$$x = r\cos\theta = 3\cos\left(\frac{4\pi}{3}\right) = -\frac{3}{2}$$
$$y = r\sin\theta = 3\sin\left(\frac{4\pi}{3}\right) = -\frac{3\sqrt{3}}{2}$$

(b) **(5점)**

$$r = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + 2^2} = \sqrt{13}$$
$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(-\frac{2}{3}\right) = -\tan^{-1}\left(\frac{2}{3}\right)$$

4. **(총 10점, 각 5점)**

(a) **(5점)**

$$x = r\sin\theta\cos\phi = 2 \times \sin\left(\frac{2\pi}{3}\right) \times \cos\left(\frac{3\pi}{4}\right) = 2 \times \frac{\sqrt{3}}{2} \times -\frac{\sqrt{2}}{2} = -\frac{\sqrt{6}}{2}$$
$$y = r\sin\theta\sin\phi = 10 \times \sin\left(\frac{2\pi}{3}\right) \times \sin\left(\frac{3\pi}{4}\right) = 2 \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{2}$$
$$z = r\cos\theta = 2 \times \cos\left(\frac{2\pi}{3}\right) = 2 \times -\frac{1}{2} = -1$$

(b) **(5점)**

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + 2^2 + 2^2} = 3$$
$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{2}{3}\right)$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{2}{1}\right) = \tan^{-1} 2$$

5. **(총 20점)**

(a) **(5점)** 문제의 (1) 식을 다시 쓰면

$$\psi_{322}(r,\theta,\phi) = \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin^2\theta e^{2i\phi}$$
$$= \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin^2\theta\big(\cos 2\phi + i\sin 2\phi\big) \quad (2)$$

그러므로

$$\mathrm{Re}(\psi_{322}) = \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin^2\theta\cos 2\phi,$$
$$\mathrm{Im}(\psi_{322}) = \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin^2\theta\sin 2\phi, \quad (3)$$

(b) **(5점)**

$$|\psi_{322}(r,\theta,\phi)|^2 = \psi^*_{322}\psi_{322} = \frac{1}{162^2\pi}\left(\frac{1}{a_0}\right)^3\left(\frac{r}{a_0}\right)^4 e^{-2r/(3a_0)}\sin^4\theta \tag{4}$$

(c) **(10점)** 위의 (2) 식으로부터

$$\begin{aligned}
\psi_{322}(r,\theta,\phi) &= \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{r}{a_0}\right)^2 e^{-r/(3a_0)}\sin^2\theta e^{2i\phi} \\
&= \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{7/2} e^{-r/(3a_0)} r^2\sin^2\theta e^{2i\phi} \\
&= \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{7/2} e^{-r/(3a_0)}\big(r\sin\theta e^{i\phi}\big)^2 \\
&= \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{7/2} e^{-r/(3a_0)}\big\{r\sin\theta(\cos\phi + i\sin\phi)\big\}^2 \\
&= \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{7/2} e^{-r/(3a_0)}\big\{r\sin\theta\cos\phi + ir\sin\theta\sin\phi\big\}^2
\end{aligned} \tag{5}$$

3차원 직교 좌표계와 구면 극 좌표계의 관계식으로부터

$$\begin{aligned}
x &= r\sin\theta\cos\phi \\
y &= r\sin\theta\sin\phi \\
r &= \sqrt{x^2+y^2+z^2}
\end{aligned}$$

이므로, 관계식을 위의 (5) 식에 대입하면

$$\psi_{322}(x,y,z) = \frac{1}{162\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{7/2} e^{-\sqrt{x^2+y^2+z^2}/(3a_0)}\big(x+iy\big)^2 \tag{6}$$